

Ridho Khalis Prawira W

Software Engineer

Email: Ridhokhalis37@gmail.com | LinkedIn: [Ridho Khalis](#) | Website: [Ridhokhalis](#) | Github: [Ridhokhalis99](#)

SUMMARY

Backend Software Engineer with 4+ years of experience building scalable, high-throughput systems in the FinTech domain. Strong expertise in Ruby on Rails, Node.js, and distributed systems, with hands-on experience in fraud detection, financial integrations, and data-intensive workloads. Proven ability to optimize system performance, integrate third-party financial platforms, and deliver production-grade backend services in high-impact environments.

EXPERIENCE

Software Engineer – Jack Finance & Transfez

Aug 2022 - Present

- Built "Mbak Tari", a multi-agent WhatsApp remittance assistant (n8n, LangChain, MCP) with intent routing, RAG-based FAQ, and a deterministic hallucination checker that guarantees rates, amounts, and payment links come from tool output.
- Designed and implemented a secure transaction and KYC integration middleware with external partners, applying idempotent transaction handling to ensure reliable and consistent financial operations.
- Implemented centralized and structured logging on GCP to improve observability and enable cross-service tracing in production, improving developer efficiency in debugging backend services.
- Implemented a fraud detection system using ClickHouse Materialized Views, reducing analytical query latency by approximately 6x and enabling faster fraud analysis on high-volume transaction data.
- Built and maintained a secure accounting integration middleware to synchronize financial transactions and statements with external Odoo services, ensuring data consistency and reliability across systems.

SKILLS

Languages & Frameworks: Node.js, Ruby on Rails, Express.js, NestJS, Java (Spring Boot), Go (Gin)

AI & Machine Learning: LLM agents (LangChain, LangGraph, n8n), RAG, tool calling / MCP, pgvector, LightGBM, survival analysis, prompt evaluation

Databases & Data Stores: PostgreSQL, MySQL, ClickHouse, Redis, BullMQ, Kafka

API & Architecture: RESTful APIs, GraphQL, Microservices Architecture, System Design

Cloud & DevOps: AWS, Docker, Kubernetes, Git, Firebase, Kafka

Testing & Performance: Jest, Mocha, k6, Sentry, Logging & Monitoring

CERTIFICATION

AWS Certified Solutions Architect Associate (SAA-C03) – Amazon

Dec 2025 – Dec 2028

Web Services

- Demonstrated knowledge in designing secure, scalable, and highly available architectures on AWS.

- Experienced in selecting appropriate AWS services for compute, storage, networking, and databases based on workload requirements.
- Applied best practices for cost optimization, security, and reliability in cloud-based system design.

PROJECTS

Rekanka

Sep 2026

Multi-tenant operating system for small F&B businesses: full POS, recipe costing, stock and batch production, and GoFood/GrabFood/ShopeeFood settlement reconciliation into profit per channel. Tenant isolation is enforced with PostgreSQL row-level security. The PWA has role-based work modes (owner, cashier, kitchen) plus an AI assistant that proposes changes for human confirmation, covered by Playwright end-to-end tests.

Techstack: NestJS, Next.js, PostgreSQL (Drizzle, RLS), Redis, BullMQ, MCP, Docker

Koogita

Aug 2026

AI journaling app with an async BullMQ pipeline that transcribes voice notes, embeds them into pgvector, and extracts emotions, thinking patterns, and topics as structured LLM output. Includes retrieval-based Q&A across a user's notes and a LangGraph workflow for multi-perspective analysis, with per-model cost tracking and usage quotas.

Techstack: NestJS, Next.js, PostgreSQL (pgvector, Prisma), Redis, BullMQ, LangGraph

Ledgers

Mar 2026

Distributed double-entry ledger system for handling high-throughput financial transactions with strong consistency guarantees. The system adopts a CQRS-style architecture by separating write and read services, reducing contention while ensuring consistency through idempotent transactions and serializable database operations. Deployed on Kubernetes with CPU-based Horizontal Pod Autoscaling (HPA) and PostgreSQL managed via CloudNativePG (CNPG) with primary-replica replication, improving scalability and stability. Load testing with k6 sustained 1000 virtual users with ~47K requests, 0% error rate, and ~179ms p95 latency.

Repository: github.com/ridhokhalis99/ledgers

Techstack: Go (Gin), PostgreSQL, Kubernetes, CNPG, Docker, k6

Link Shortener

Jun 2025

Microservice-based application for generating and tracking shortened URLs. The main service handles URL creation and redirection, while the analytics service consumes real-time events such as IP address, user agent, referrer, and timestamp using Kafka for decoupled communication.

Link Shortener repository: github.com/ridhokhalis99/linkshortener

Analytics repository: github.com/ridhokhalis99/linkshortener-analytics

Kubernetes repository: github.com/ridhokhalis99/linkshortener-k8s

Techstack: Java, Spring Boot, Kafka, PostgreSQL, Docker, Kubernetes

EDUCATION

Full Stack Javascript - Hacktiv8

Mar – July 2022

Score: 97.4/100

Bachelor of Nuclear Engineering - Universitas Gadjah Mada

Aug 2017 – Mar 2022

GPA: 3.14/4.00